

Adoption of EVs will accelerate only when the batteries in the vehicles become safer, and charging infrastructure and battery swapping stations become readily available. Renon's latest product, Groot, is a swappable smart battery pack designed especially for electric 2-wheelers running on Indian road conditions.

"At Renon we give a lot of importance to cell qualification, characterisation, and cell testing," says Ganesh Moorthi, CTO of Renon. He adds, "Selection of cell is crucial for both the performance and safety of the battery pack." Renon batteries conform to the United Nations standard UN38.3 for lithium batteries, which involves mechanical drop, impact crush, freefall, nail penetration, overcharge, post-discharge, etc.

Groot uses 18650 type lithium nickel manganese cobalt oxide (NMC) cells due to their high energy density. The removable battery



Groot, the swappable smart battery pack

pack is rated for 2.04kWh and has a nominal voltage of 57.5V. Groot weighs around 10kg, which makes it easier to handle without any special equipment. The smart battery pack employs a smart indigenous BMS that offers multiple protection features and adds telematics for improved real-time data sharing.

Talking about

the BMS, Moorthi says, "BMS is important for the safety of the battery pack. It tracks the different states of the battery pack, such as the state of charge, the state of the energy state of power, the state of health, etc. The right BMS can help us extract the highest efficiency and performance from the battery packs."

Groot offers multiple levels of thermal management and runaway mitigation solutions, which makes it safer to employ in new and existing 2-wheelers driven in hot Indian climate. The battery pack also employs a screw-type pressure relief valve, which further improves the safety of the swappable battery pack during an extended period of use.

Renon India has seen massive growth since it started in 2019. The company plans to reach a 500 + MWh production line in the near future, producing 3000 + packs monthly. They aim to generate a revenue of 400 million rupees in current financial year, up from 250 million rupees in 2021. They expect to grow their revenue at a CAGR of 50% during the next three years and want to become a leader in all the verticals of energy storage space. **EFY**

—Sharad Bhowmick

Advantages of battery swapping

- Lowers cost of EV (as battery belongs to swapping station)
- Improves battery life
- Lowers charging time for the customer
- Lowers maintenance cost of EV
- Improves safety of the EV (as swapping station keeps check on battery's health)

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